Transferability estimation

Which pre-trained model should I fine-tune, and can I tell without fine-tuning them all?

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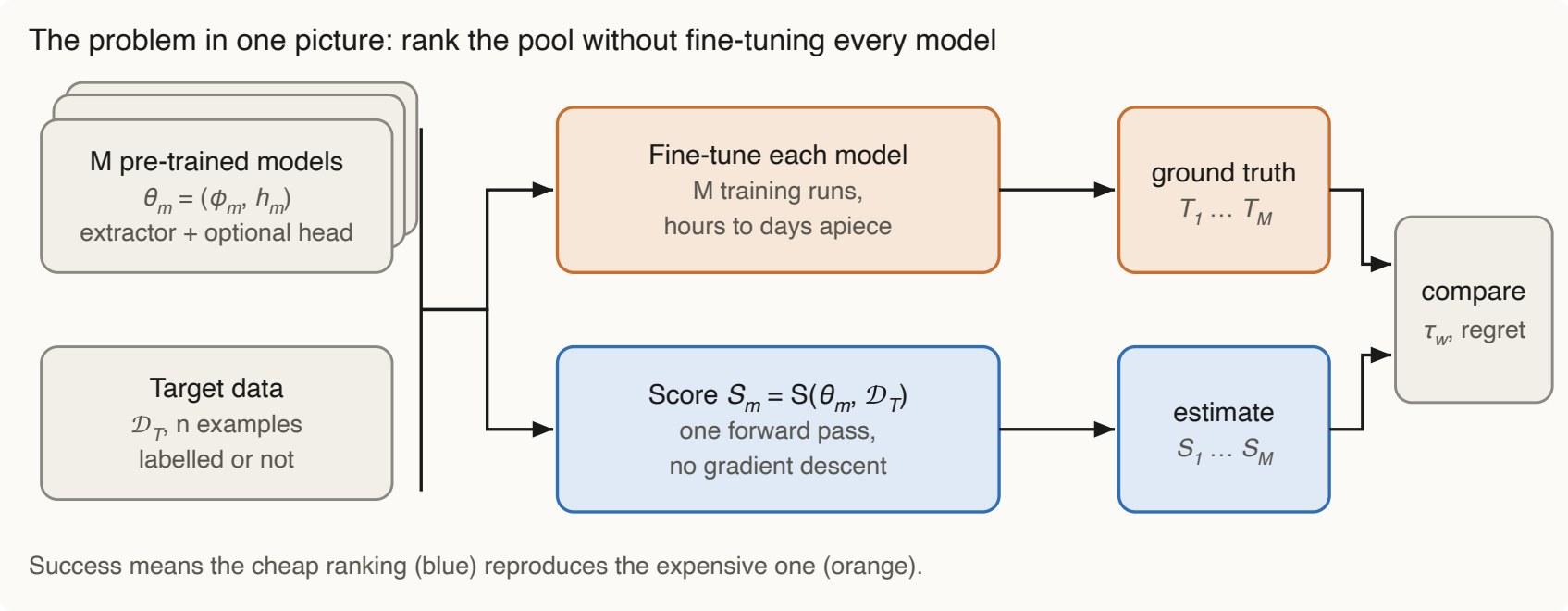
The problem · six scores from NCE to PAS · two benchmarks

Built from the annotated notes in `theory_inclined_papers_with_annotations`.

The practical question

A model zoo holds M pre-trained models. You have one new target task with n examples.

- **Gold standard:** fine-tune all M, evaluate, keep the best. Cost: M training runs, each with its own hyper-parameter search
- **Transferability estimation:** compute a cheap score S_m per model, from the frozen model and the target data, and take the top one



The problem, in symbols

Given

- a pool $\theta_m = (\phi_m, h_m), m = 1, \ldots, M$: feature extractor $\phi_m : \mathcal{X} \to \mathbb{R}^{D_m}$, optional head h_m from a source task $\mathcal{T}_m^S$ (its data $\mathcal{D}_m^S$ may be unavailable)
- target data $\mathcal{D}^T$ of n examples, and a test distribution P^T

Ground truth (expensive): $T_m = \mathrm{perf}\left(\mathrm{FT}(\theta_m; \mathcal{D}^T);\ P^T\right)$, the accuracy, balanced accuracy, AUROC or log-likelihood after fine-tuning (or after retraining only the head)

Estimator (cheap): $S_m = S(\theta_m; \mathcal{D}^T)$, one forward pass, no gradient descent

Goal: make $\tau_w\big((S_m)_m, (T_m)_m\big)$ high, or, for pure selection, make the regret $\max_m T_m - T_{\hat{m}}$ small, where $\hat{m} = \arg\max_m S_m$

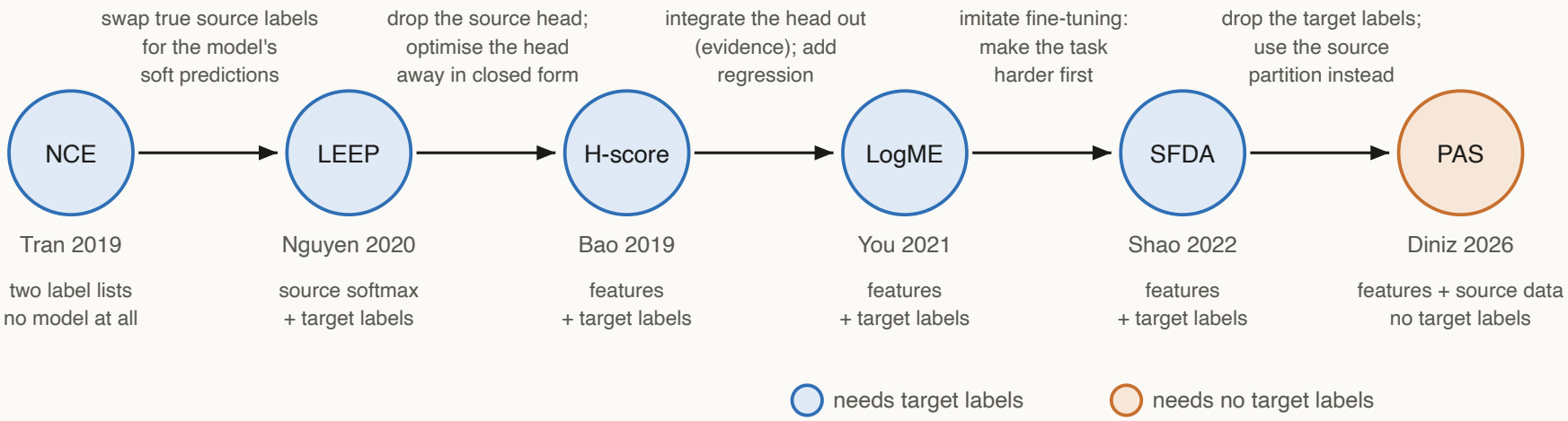
How a score is judged

- Kendall's $\tau = \dfrac{2}{M(M-1)} \sum_{i<j} \operatorname{sgn}(T_i - T_j) \operatorname{sgn}(S_i - S_j)$: pairs ordered the same way minus pairs ordered oppositely
- If the correlation is τ, then $\Pr(T_i > T_j \mid S_i > S_j) = \frac{\tau+1}{2}$, so $\tau_w \approx 0.5$ means about 75% of pairs ordered right (LogME paper)
- **Weighted** τ_w counts a pair more when T_i, T_j, S_i, S_j are large, because you deploy the top-ranked model
- Also used: Spearman ρ, Pearson r, the regret of the top pick

Notes: You 2021 — LogME (the τ_w definition).

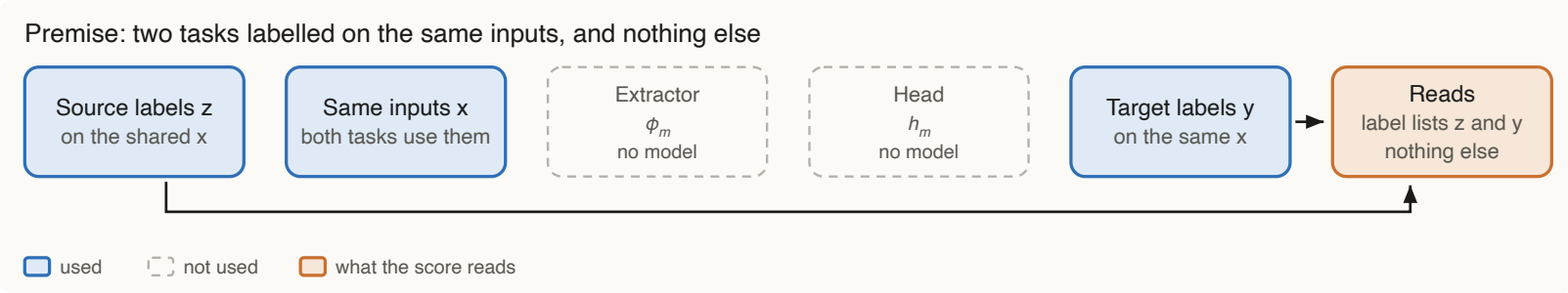
The progression we will follow

From the tractable case to the label-free one: what changes at each step



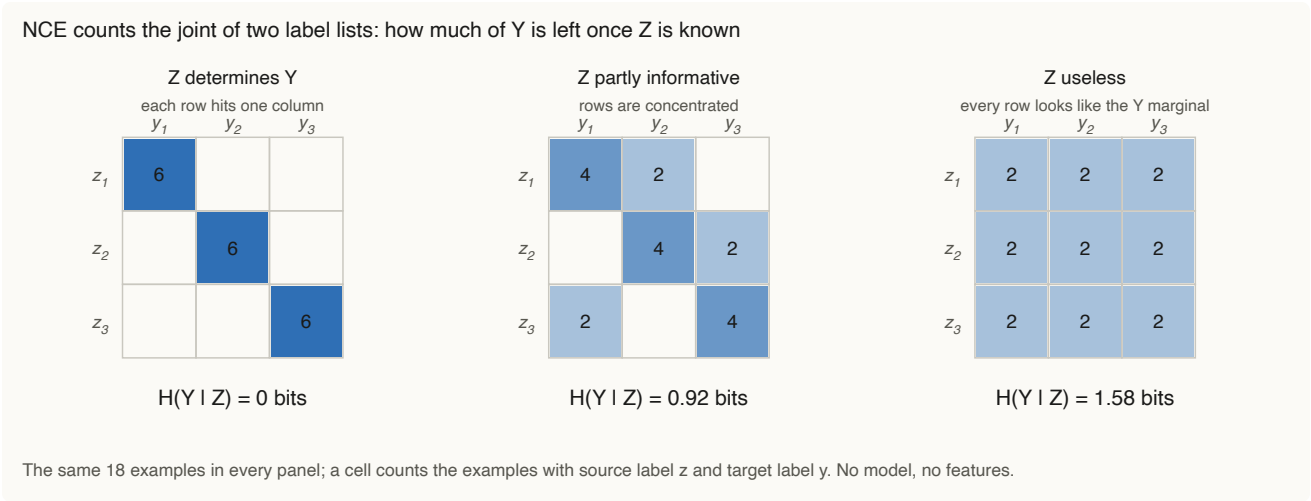
A conceptual order, not a chronological one: NCE and H-score are both 2019, LEEP 2020, LogME 2021, SFDA 2022, PAS 2026.

NCE (Tran 2019): the tractable case

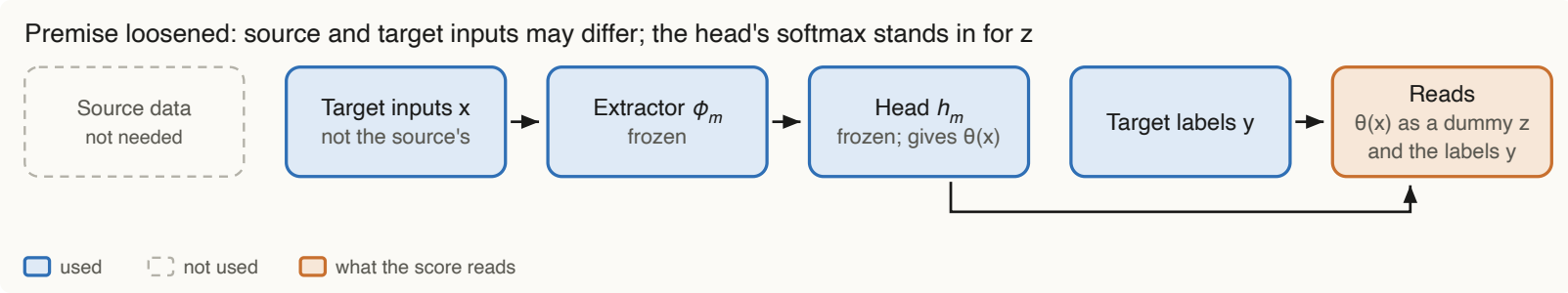


Two tasks on the **same inputs**, with ground-truth label sequences Z (source) and Y (target). No model, no features, no input data: count the joint $\hat{P}(y, z)$ and compute

$$H(Y \mid Z) = -\sum_{y,z} \hat{P}(y, z) \log \hat{P}(y \mid z)$$

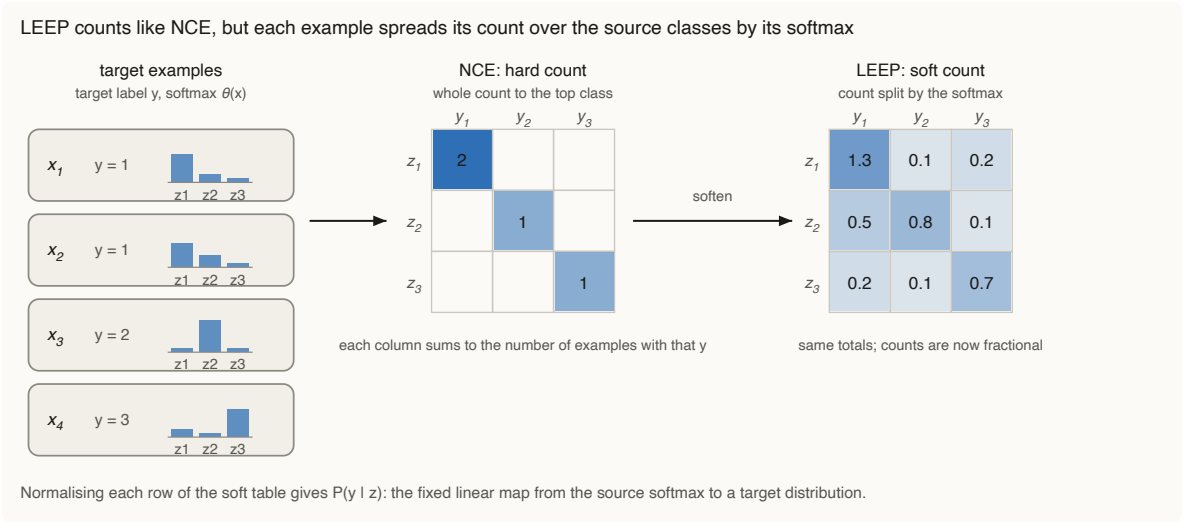


LEEP (Nguyen 2020): use the model's softmax instead



Push target inputs through the frozen source classifier: $\theta(x_i) \in \Delta(\mathcal{Z})$, a "dummy" label distribution, and count softly:

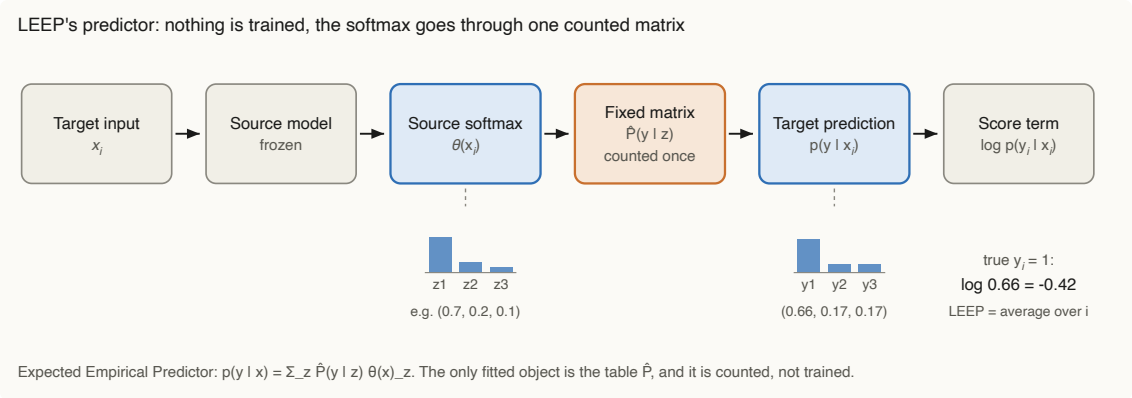
$$\hat{P}(y, z) = \frac{1}{n} \sum_{i:\, y_i = y} \theta(x_i)_z, \qquad \hat{P}(y \mid z) = \frac{\hat{P}(y, z)}{\hat{P}(z)}$$



LEEP: the Expected Empirical Predictor

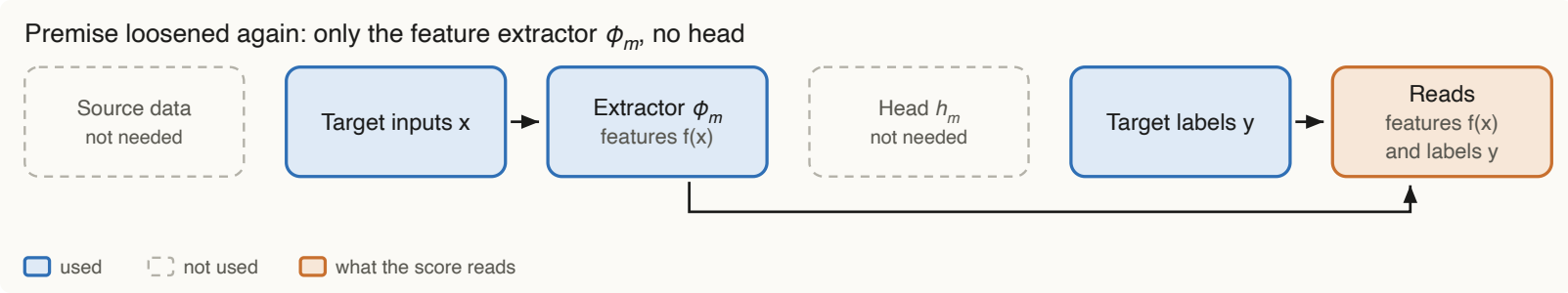
The **Expected Empirical Predictor** is a fixed linear map of the softmax, $p(\cdot \mid x) = \hat{P}^{\top}\theta(x)$, and

$$\mathrm{LEEP} = \frac{1}{n}\sum_{i=1}^{n} \log \sum_{z \in \mathcal{Z}} \hat{P}(y_i \mid z)\, \theta(x_i)_z$$



Notes: Nguyen 2020 — LEEP

H-score (Bao 2019): solve the head, then eliminate it



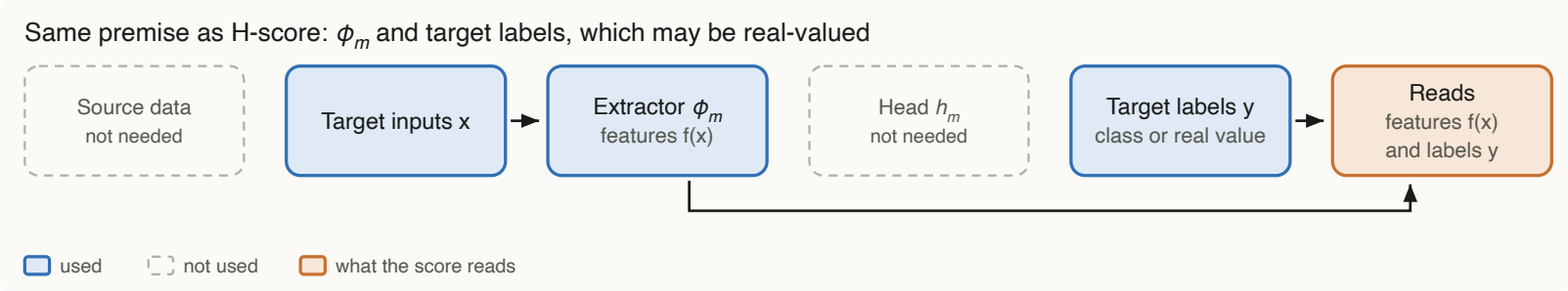
Frozen features $f(x) \in \mathbb{R}^k$, a linear head, target labels.

$$\mathcal{H}(f) = \operatorname{tr}\left(\operatorname{cov}\left(f(X)\right)^{-1} \operatorname{cov}\left(\mathbb{E}[f(X) \mid Y]\right)\right)$$



Notes: Bao 2019 — H-score

LogME (You 2021): integrate the head out



A Bayesian linear model on frozen features $F \in \mathbb{R}^{n \times D}$. Score the **evidence**, not the likelihood at the best w^*:

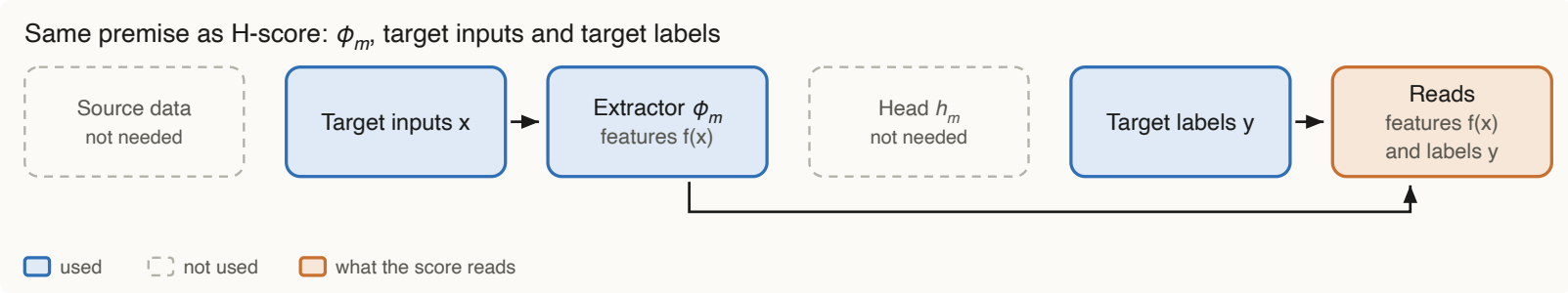
$$p(y \mid F, \alpha, \beta) = \int p(w \mid \alpha)\, p(y \mid F, w, \beta)\, \mathrm{d}w$$

$$\mathrm{LogME} = \frac{1}{n}\left[\frac{n}{2}\ln\beta^* + \frac{D}{2}\ln\alpha^* - \frac{n}{2}\ln 2\pi - \frac{1}{2}\ln|A| - \frac{\beta^*}{2}\|Fm - y\|^2 - \frac{\alpha^*}{2}m^\top m\right]$$

with $A = \alpha^* I + \beta^* F^\top F$, $m = \beta^* A^{-1} F^\top y$, and α^*, β^* from a fixed point.

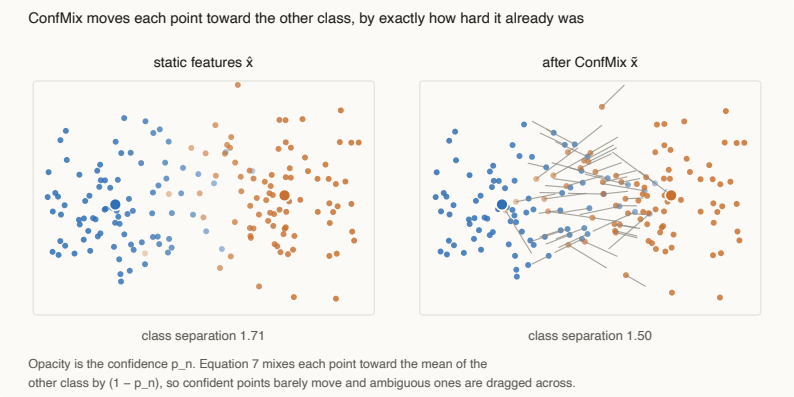
Notes: You 2021 — LogME

SFDA (Shao 2022): imitate fine-tuning, not only the features



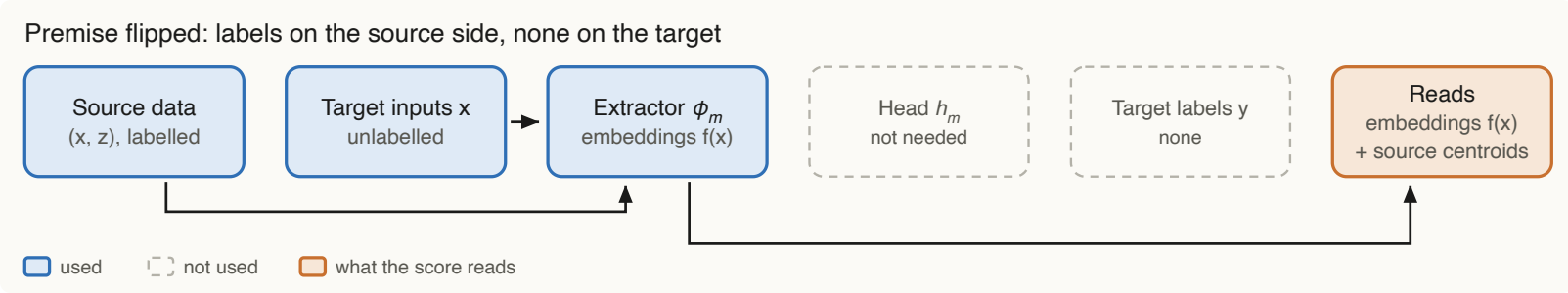
Fine-tuning separates classes and learns hard examples late, so a static score should imitate both.

- **Reg-FDA:** a regularised Fisher discriminant projects into $C - 1$ dimensions, LDA classifies there, each sample gets a confidence p_n
- **ConfMix:** moves each sample toward the other classes by its own difficulty. Re-run Reg-FDA; the score is the average log-likelihood of the second pass



Notes: Shao 2022 — SFDA

PAS (Diniz 2026): no target labels at all



Labelled source, unlabelled target. Unit-norm embeddings, source class centroids $\mu_c = \dfrac{\sum f_\theta(x_i^S)}{\left\|\sum f_\theta(x_i^S)\right\|}$, cosine distance $1 - f_\theta(x) \cdot \mu_c$. With d_1, d_2 the nearest and second-nearest centroid distances,

$$\mathrm{PAS} = \frac{1}{|\mathcal{D}^T|} \sum_j \frac{d_{2j} - d_{1j}}{d_{2j}} = 1 - \mathbb{E}\left[\frac{d_1}{d_2}\right]$$

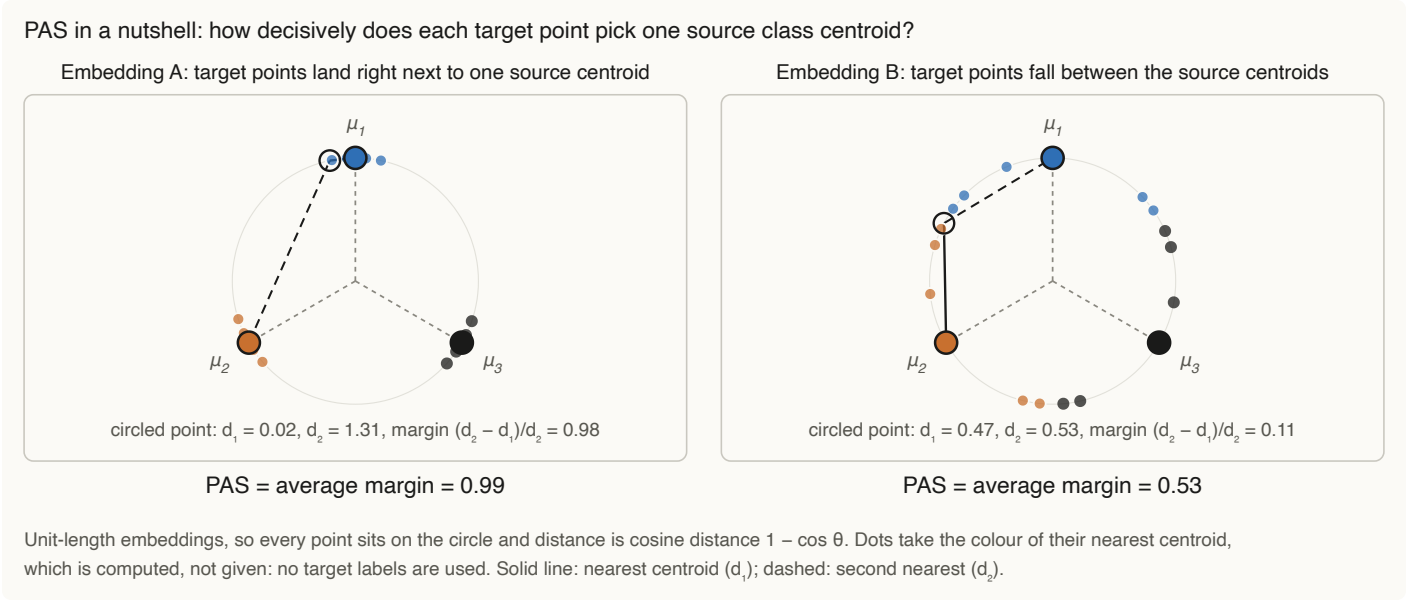
- A Silhouette score with "own cluster" replaced by "nearest cluster": the **average relative margin** of a nearest-centroid classifier, the complement of Lowe's ratio test
- Asymmetric by design. Range $[0, 1]$

Notes: Diniz 2026 — PAS

PAS: the core idea

PAS uses the **source** classes as a ruler and asks how decisively each unlabelled target point picks one of them:

1. embed to unit length and average each source class into a centroid μ_c
2. for each target point take the cosine distances to all centroids; d_1 and d_2 are the smallest and second smallest
3. its margin is $(d_2 - d_1)/d_2$: 1 on a centroid, 0 when equally close to two. PAS is the average margin

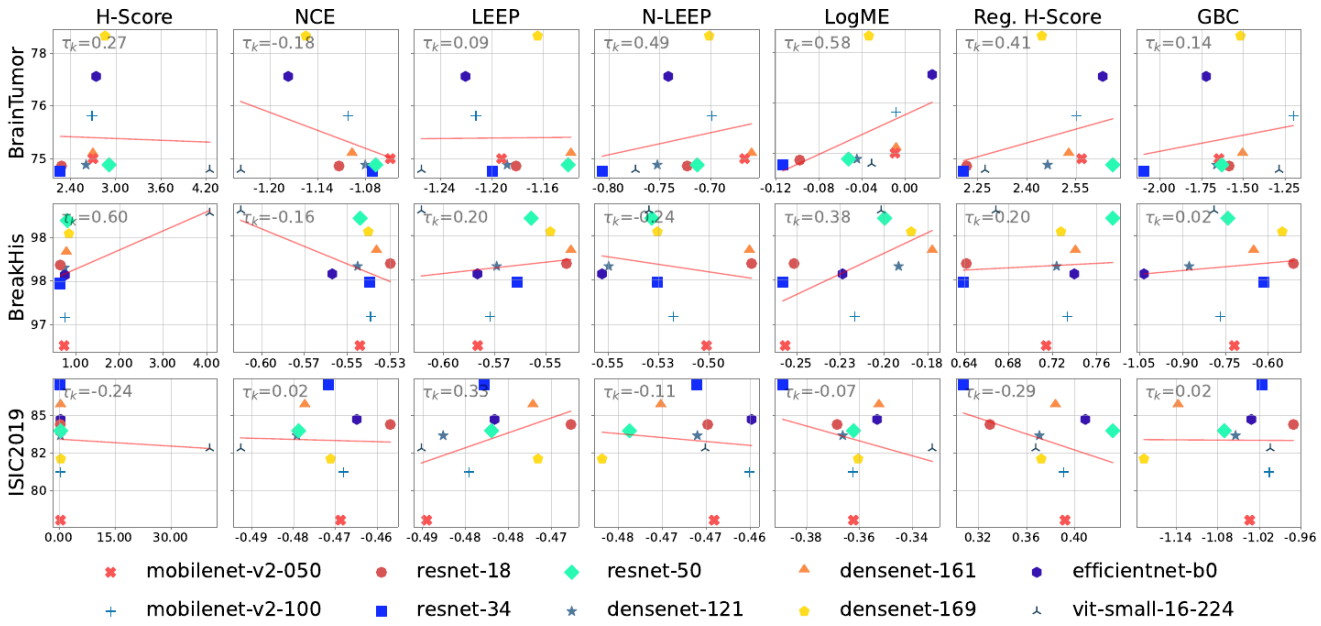


Notes: Diniz 2026 — PAS

Gaps

Do these scores survive medical imaging?

Chaves 2023: seven scores, ten models, three medical tasks



- **In-distribution:** unstable

Figure: Chaves et al. 2023, Fig. 3 (arXiv:2308.07444). Notes: Chaves 2023 — medical TE

Claßen 2026: do rankings survive a change of seed?

targets, sources	8 MedMNIST 2D sets; the other MedMNIST sets plus ImageNet
model	ResNet-18 only
metrics	H-score, LEEP, $\mathcal{N}$LEEP, LogME, NCTI, PARC, SFDA
design	nested stratified subsets of 5–75%, 5 seeds; reference tuned for accuracy and for AUROC

1. **Rankings move under nothing but the seed.** On Breast LogME ranks Blood as both the best and the worst source
2. **The reference ranking moves too.** Accuracy against AUROC reorders almost every source; one goes from 2nd to 11th

Also: $\mathcal{N}$LEEP and SFDA produce NaNs.

Notes: Claßen 2026 — TE robustness

Explorations

Mode connectivity as a transferability score

To do: mode connectivity between source and target

Question. Can the loss landscape between the source solution and the fine-tuned target solution explain, and predict, transferability?

- **Endpoints:** the source model $\theta_S = \theta_m$ from the pool, and the fine-tuned target model $\theta_T^* = \mathrm{FT}(\theta_m; \mathcal{D}^T)$, the **oracle** that defines T_m
- **Landscape:** the target loss $\mathcal{L}_T(\theta)$, and the source loss $\mathcal{L}_S(\theta)$ for contrast
- **Path:** $\gamma : [0, 1] \to \Theta$ with $\gamma(0) = \theta_S$ and $\gamma(1) = \theta_T^*$: a straight line, or a learned curve (Bézier or polygonal chain) that keeps the loss low, as in the mode-connectivity literature
- **Model:** fine-tuning as motion through this landscape. The height, length and bending of the path say how far the source solution must be reshaped for the target task

Related notes: Achille 2019 — task reachability (fine-tuning as a dynamical process)

To do: a path score as a transferability metric

Characterise the path by functionals of $\mathcal{L}_T$ along it:

$$\mathcal{I}_T = \int_0^1 \mathcal{L}_T\big(\gamma(t)\big)\,\|\dot\gamma(t)\|\,\mathrm{d}t, \qquad B = \max_t \Big[\mathcal{L}_T\big(\gamma(t)\big) - (1-t)\mathcal{L}_T(\theta_S) - t\,\mathcal{L}_T(\theta_T^*)\Big], \qquad \ell = \int_0^1 \|\dot\gamma(t)\|\,\mathrm{d}t$$

1. **Oracle runs:** fine-tune θ_T^* per source model, trace γ and record $\mathcal{I}_T$, B, ℓ and the curvature (Hessian or Fisher along the path)
2. **Explain:** which functional tracks the ground truth T_m, by τ_w and regret, against H-score, LogME, SFDA and the rest?
3. **Decode:** find a cheap proxy $\hat{S}_m$ for the winning functional that needs no fine-tuning (from features, Fisher spectrum or a few gradient steps), and calibrate it to the oracle
4. **Test:** rankings stable across seeds (Claßen 2026) and on the medical benchmarks (Chaves 2023)?

Open: a pre-trained and a fine-tuned model often share a basin, so the barrier may be near zero and the signal may sit in path length and curvature; permutation symmetry needs care when the endpoints differ more.

Sources

Notes (rendered): `msrepo.github.io/theory_inclined_papers_with_annotations/` followed by

`2019-tran-nce-hardness` · `2020-nguyen-leep` · `2019-bao-hscore-transferability` · `2021-you-logme` · `2022-shao-sfda` · `2026-diniz-pas` · `2023-chaves-medical-transferability` · `2026-classen-te-robustness`

and for the other axes: `2026-wang-ntk-selector`

Papers: Tran, Nguyen & Hassner, ICCV 2019, arXiv:1908.08142 · Nguyen et al., ICML 2020, arXiv:2002.12462 · Bao et al., ICIP 2019, arXiv:2212.10082 · You et al., ICML 2021, arXiv:2102.11005 · Shao et al., ECCV 2022, arXiv:2207.03036 · Diniz et al., ICLR 2026, arXiv:2604.09863 · Chaves et al., arXiv:2308.07444 · Claßen et al., arXiv:2608.09999

Supplementary

H-score: between-class covariance

For C classes with priors π_c and class means $\mu_c = \mathbb{E}[f(X) \mid Y = c]$:

$$\Sigma_B = \operatorname{cov}\left(\mathbb{E}[f(X) \mid Y]\right) = \sum_{c=1}^{C} \pi_c \left(\mu_c - \bar{\mu}\right)\left(\mu_c - \bar{\mu}\right)^\top, \qquad \bar{\mu} = \sum_c \pi_c \mu_c$$

From data, with n_c of the n examples in class c: $\hat{\Sigma}_B = \sum_c \frac{n_c}{n} (\hat{\mu}_c - \hat{\mu})(\hat{\mu}_c - \hat{\mu})^\top$.

- **Two classes:** $\Sigma_B = \pi_1 \pi_2 \left(\mu_1 - \mu_2\right)\left(\mu_1 - \mu_2\right)^\top$, rank one, pointing along the line joining the centroids
- **Along a direction** w: $w^\top \Sigma_B w$ is the variance of the class means projected on w: the class gap, with the within-class scatter left out
- **Total variance splits:** $\Sigma = \Sigma_W + \Sigma_B$, so $\operatorname{tr}\left(\Sigma^{-1}\Sigma_B\right)$ is the share of the spread the labels explain, summed over directions

H-score worked example: by hand

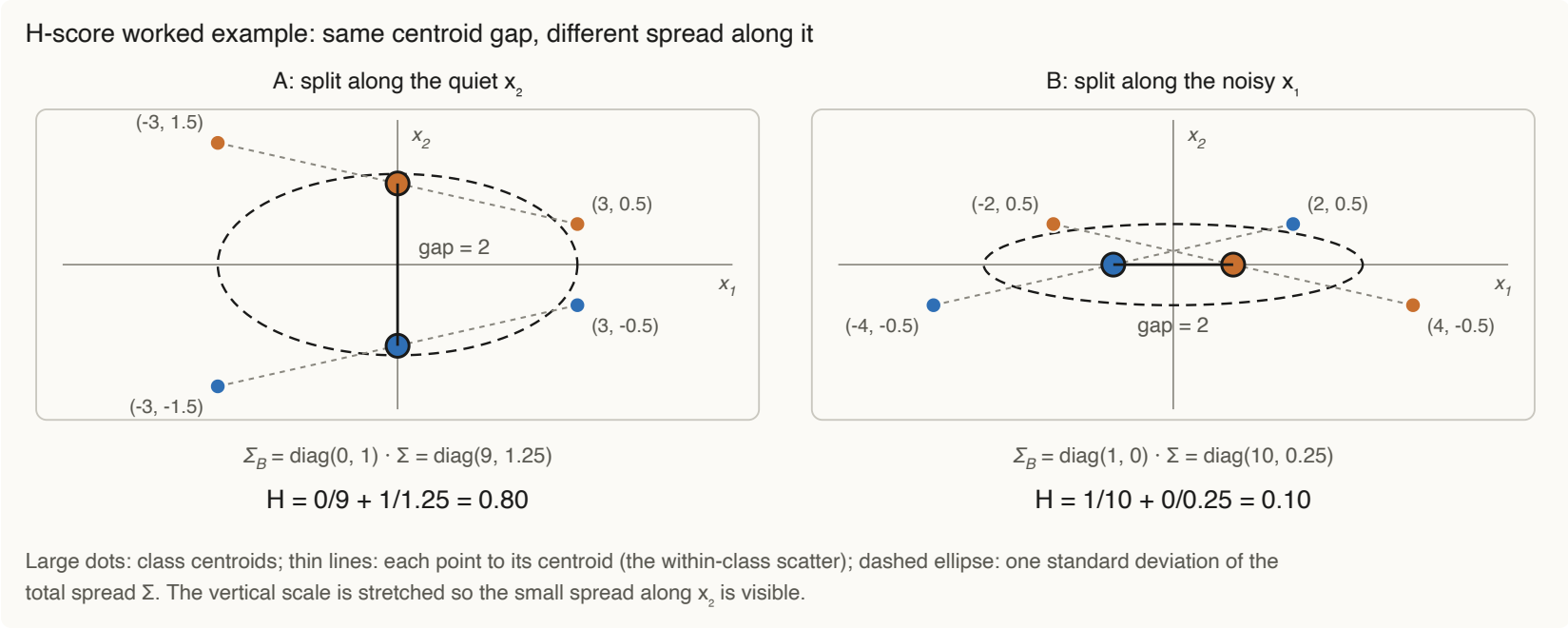
Two classes, equal priors, 2-D features (x_1, x_2), four points each case. Same centroid gap of 2.

	A: split along the quiet x_2	B: split along the noisy x_1
class 0	$(-3, -1.5),\ (3, -0.5)$	$(-4, -0.5),\ (2, 0.5)$
class 1	$(-3, 1.5),\ (3, 0.5)$	$(-2, 0.5),\ (4, -0.5)$
means	$\mu_0 = (0, -1),\ \mu_1 = (0, 1)$	$\mu_0 = (-1, 0),\ \mu_1 = (1, 0)$
Σ_B	$\mathrm{diag}(0,\ 1)$	$\mathrm{diag}(1,\ 0)$
Σ	$\mathrm{diag}(9,\ 1.25)$	$\mathrm{diag}(10,\ 0.25)$
$\Sigma^{-1}\Sigma_B$	$\mathrm{diag}(0,\ 0.8)$	$\mathrm{diag}(0.1,\ 0)$
$\mathcal{H} = \mathrm{tr}$	**0.80**	**0.10**

The raw gap is the same, so $\mathrm{tr}\,\Sigma_B = 1$ in both. In B each point is already scattered by ± 3 along x_1 within its class, so a gap of 2 is small against the spread; in A the spread along x_2 is about 1.

Notes: Bao 2019 — H-score

H-score worked example: the picture



Notes: Bao 2019 — H-score

LogME worked example: by hand

LogME integrates the weights out, so the evidence is a Gaussian density at the target: $y \sim \mathcal{N}(0, C),\ C = FF^\top/\alpha + I/\beta$, and $\mathrm{LogME} = \max_{\alpha,\beta} \frac{1}{n} \log p(y \mid F, \alpha, \beta)$. Here $\alpha = \beta = 1$ is fixed (the real score tunes both), with $n = 2$ examples, one feature and target $y = (1, -1)^\top$.

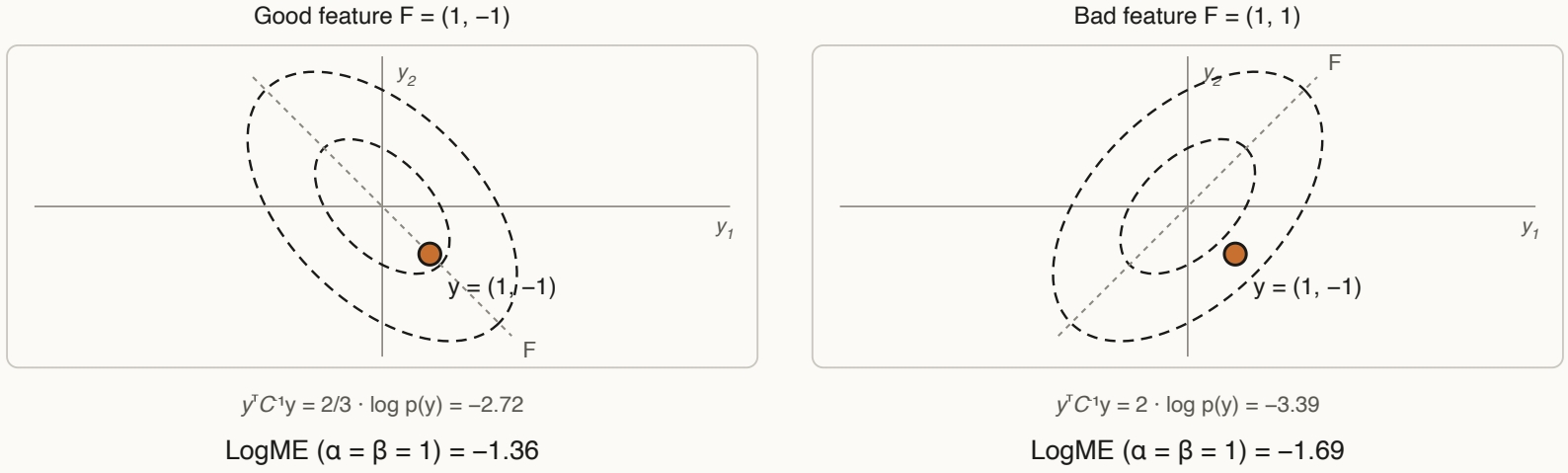
	good feature	bad feature
F	$(1, -1)^\top$, aligned with y	$(1, 1)^\top$, orthogonal to y
$C = FF^\top + I$	$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$	$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$
$\det C$	3	3
$y^\top C^{-1} y$	$2/3$	2
$\log p(y) = -\frac{1}{2}\left[n \ln 2\pi + \ln \det C + y^\top C^{-1} y\right]$	-2.72	-3.39
$\mathrm{LogME} = \log p(y)/n$	-1.36	-1.69

The determinant (the "Occam" cost) is the same, so the whole gap is the data-fit term $y^\top C^{-1} y$: y is a typical draw of $\mathcal{N}(0, C)$ only when F points along it.

Notes: You 2021 — LogME

LogME worked example: the picture

LogME worked example: where the target y falls under the evidence N(0, C), C = FF$^\intercal$ + I



Dashed rings: 1 and 2 standard deviations of the evidence N(0, C); thin line: the feature direction F, the long axis of the ring (variance 3; the short axis has variance 1). y sits inside the 1σ ring when F is aligned with it, and outside it otherwise.

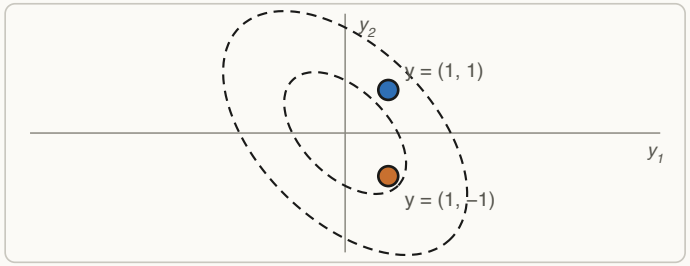
Notes: You 2021 — LogME

LogME: evidence versus likelihood

The likelihood at the best weights asks what the model *could* do; the evidence $\int p(w \mid \alpha)\, p(y \mid F, w, \beta)\, \mathrm{d}w$ asks how probable y was under the model *before* seeing it. A rigid and a flexible model can fit y equally well and still differ.

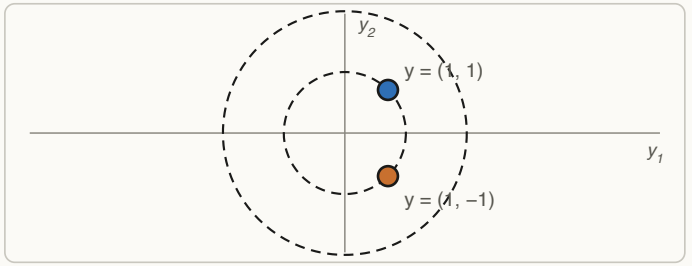
LogME: the same best-case fit, but different evidence (α = β = 1, n = 2)

Model A, rigid: one feature F = (1, −1)



	best-case likelihood	evidence
y = (1, −1)	−1.84	−2.72
y = (1, 1)	−2.84	−3.39

Model B, flexible: two features F = I



	best-case likelihood	evidence
y = (1, −1)	−1.84	−3.03
y = (1, 1)	−1.84	−3.03

Dashed: 1 and 2 standard deviations of each model's evidence density N(0, C). For y = (1, −1) the likelihood ties, because both models fit it exactly; the evidence prefers the rigid model, which predicted it. For y = (1, 1) it prefers B.

Notes: You 2021 — LogME

LogME: step 1, writing the two Gaussians

A D-dimensional Gaussian with isotropic precision is a constant times the exponential of a squared length:

$$\mathcal{N}(w; 0, \alpha^{-1} I) = \left(\frac{\alpha}{2\pi}\right)^{D/2} e^{-\frac{\alpha}{2}\|w\|^2} \qquad \mathcal{N}(y; Fw, \beta^{-1} I) = \left(\frac{\beta}{2\pi}\right)^{n/2} e^{-\frac{\beta}{2}\|y - Fw\|^2}$$

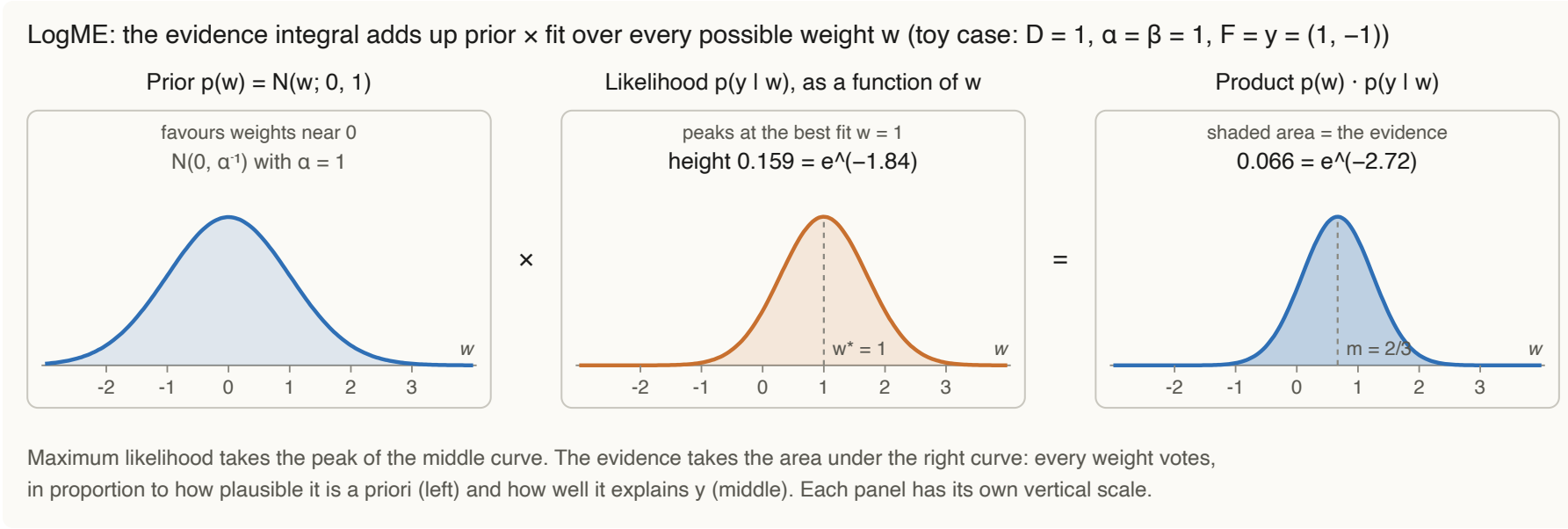
Multiplying them adds the exponents, so the product is the two constants times $e^{-E(w)}$:

$$E(w) = \frac{\beta}{2}\|Fw - y\|^2 + \frac{\alpha}{2}\|w\|^2$$

- **Prior** $\mathcal{N}(w; 0, \alpha^{-1} I)$: which weights are plausible before seeing y (small ones, if α is large)
- **Likelihood** $\mathcal{N}(y; Fw, \beta^{-1} I)$: how well *this* w explains y. As a function of w it is a bump around the best-fit weights
- **Their product** says how plausible w is *and* how well it fits; $E(w)$ is the cost of w, small for good weights
- **The evidence** $p(y)$ is the area under that product: the sum over every w, not the value at the best one

Notes: You 2021 — LogME

LogME: what the evidence integral does



Notes: You 2021 — LogME

LogME: deriving the closed-form evidence

Integrating out w multiplies two Gaussians, $p(y) = \int \mathcal{N}(w; 0, \alpha^{-1} I)\, \mathcal{N}(y; Fw, \beta^{-1} I)\, \mathrm{d}w$, which gives

$$p(y) = \left(\frac{\alpha}{2\pi}\right)^{D/2} \left(\frac{\beta}{2\pi}\right)^{n/2} \int e^{-E(w)}\, \mathrm{d}w, \qquad E(w) = \frac{\beta}{2}\|Fw - y\|^2 + \frac{\alpha}{2}\|w\|^2$$

E is quadratic in w, so **complete the square** with $A = \alpha I + \beta F^\top F$:

$$E(w) = \tfrac{1}{2}(w - m)^\top A\,(w - m) + E(m), \qquad m = \beta A^{-1} F^\top y$$

- m minimises E: the posterior mean of w. $E(m) = \frac{\beta}{2}\|Fm - y\|^2 + \frac{\alpha}{2} m^\top m$ is just a number
- The remaining integral is exact, with no approximation: $\int e^{-\frac{1}{2}(w-m)^\top A (w-m)} \mathrm{d}w = (2\pi)^{D/2} |A|^{-1/2}$

Notes: You 2021 — LogME

LogME: reading the closed-form evidence

The $(2\pi)^{D/2}$ factors cancel, and taking the log gives

$$\mathcal{L}(\alpha,\beta) = \tfrac{n}{2}\ln\beta + \tfrac{D}{2}\ln\alpha - \tfrac{n}{2}\ln 2\pi - \tfrac{1}{2}\ln|A| - \tfrac{\beta}{2}\|Fm-y\|^2 - \tfrac{\alpha}{2}m^\top m$$

term	where it comes from
$\frac{n}{2}\ln\beta,\ -\frac{n}{2}\ln 2\pi$	normaliser of the label noise
$\frac{D}{2}\ln\alpha$	normaliser of the weight prior
$-\frac{\beta}{2}\|Fm-y\|^2$	**data fit**
$-\frac{\alpha}{2}m^\top m$	**weight size**
$-\frac{1}{2}\ln\|A\|$	**Occam term** from the integral: large when F has many strong directions

Check with the worked example ($D=1, n=2, \alpha=\beta=1, F=y=(1,-1)^\top$): $A=3, m=\frac{2}{3}$, $\|Fm-y\|^2=\frac{2}{9}, m^\top m=\frac{4}{9}$, so $\mathcal{L}=-1.838-0.111-0.222-0.549=-2.72$, the same as $\log\mathcal{N}(y;0,C)$.

Notes: You 2021 — LogME

LogME: deriving the α and β updates

Maximise $\mathcal{L}(\alpha, \beta)$ by setting its derivatives to zero. Two facts make this short:

- Let λ_i be the eigenvalues of $F^\top F$. Then A has eigenvalues $\alpha + \beta\lambda_i$, so $\ln|A| = \sum_i \ln(\alpha + \beta\lambda_i)$
- m minimises E, so $\partial E / \partial m = 0$ there: only the *explicit* α, β in $\mathcal{L}$ need differentiating

$$\frac{\partial \mathcal{L}}{\partial \alpha} = \frac{D}{2\alpha} - \frac{1}{2} m^\top m - \frac{1}{2} \sum_i \frac{1}{\alpha + \beta\lambda_i} = 0 \;\Rightarrow\; m^\top m = \frac{1}{\alpha} \sum_i \frac{\beta\lambda_i}{\alpha + \beta\lambda_i} = \frac{\gamma}{\alpha}$$

$$\frac{\partial \mathcal{L}}{\partial \beta} = \frac{n}{2\beta} - \frac{1}{2} \|Fm - y\|^2 - \frac{1}{2} \sum_i \frac{\lambda_i}{\alpha + \beta\lambda_i} = 0 \;\Rightarrow\; \|Fm - y\|^2 = \frac{n - \gamma}{\beta}$$

with $\gamma = \sum_i \dfrac{\beta\lambda_i}{\alpha + \beta\lambda_i}$. Solving for the hyper-parameters:

$$\boxed{\alpha = \frac{\gamma}{m^\top m}, \qquad \beta = \frac{n - \gamma}{\|Fm - y\|^2}}$$

Notes: You 2021 — LogME

LogME: reading the fixed point

- $\gamma \in [0, D]$ **is the effective number of parameters.** Direction i counts $\frac{\beta\lambda_i}{\alpha+\beta\lambda_i} \in (0,1)$: near 1 where the data dominate the prior, near 0 where the prior does
- $\alpha = \gamma / m^\top m$**:** the prior precision at which the weight norm matches the number of well-determined parameters
- $\beta = (n-\gamma)/\|Fm - y\|^2$**:** inverse noise variance, with residual over $n-\gamma$ degrees of freedom
- **Why a fixed point:** γ and m depend on α, β, so there is no closed form. Start from any α, β, recompute γ and m, update both, repeat; then $\mathrm{LogME} = \mathcal{L}(\alpha^*, \beta^*)/n$

Check on the worked example ($\lambda = 2$, start $\alpha = \beta = 1$): $\gamma = \frac{2}{3}$, $m = \frac{2}{3}$, so $\alpha \to \frac{2/3}{4/9} = 1.5$ and $\beta \to \frac{4/3}{2/9} = 6$. The evidence rises from -2.72 to -1.81 in one step. Here y lies exactly on F, so β keeps growing; real labels have noise and the iteration settles.

Notes: You 2021 — LogME

SFDA: the Fisher discriminant (Reg-FDA)

Find the projection where the classes are far apart relative to their spread. For one direction u, with between- and within-class scatter S_B and S_W:

$$J(u) = \frac{u^\top S_B u}{u^\top S_W u}$$

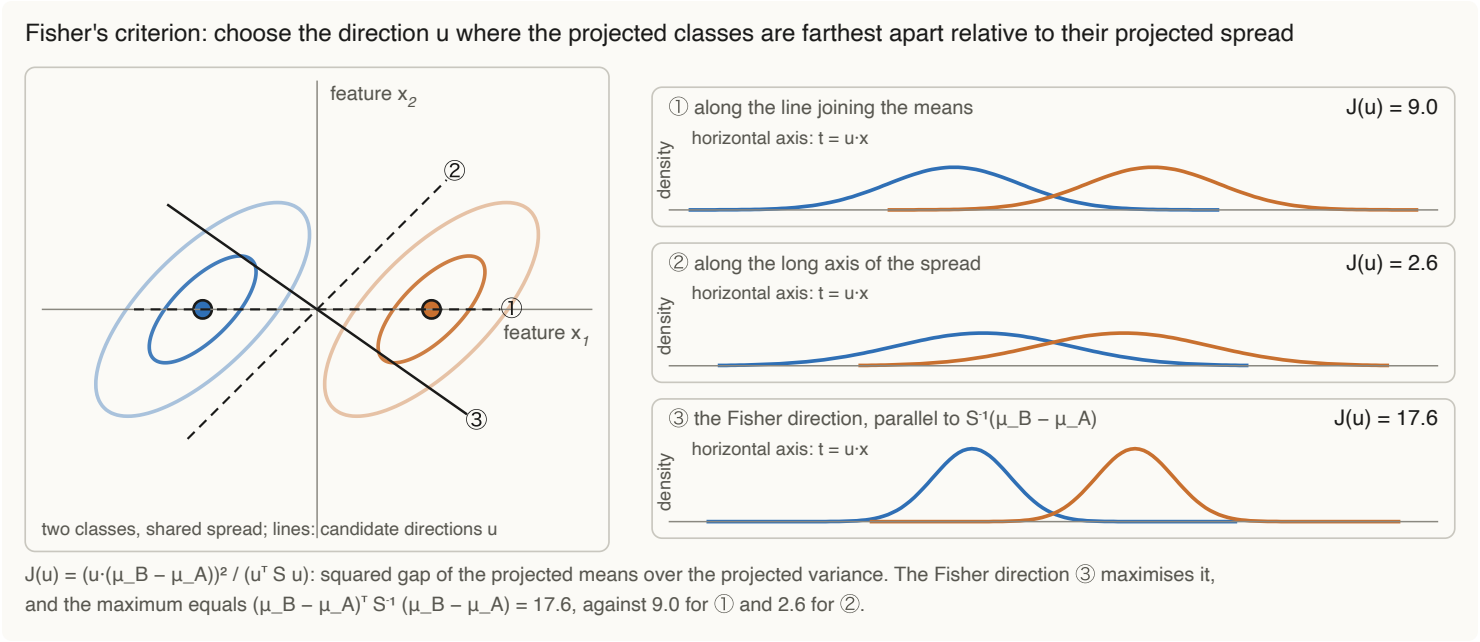
J is unchanged by rescaling u, so maximise $u^\top S_B u$ subject to $u^\top S_W u = 1$. The Lagrange condition is a **generalised eigenproblem**:

$$S_B\, u = v\, S_W\, u$$

- **Several directions:** SFDA maximises $\dfrac{|U^\top S_B U|}{|U^\top \tilde{S}_W U|}$; the solution is the top eigenvectors of $\tilde{S}_W^{-1} S_B$, and the maximum is the product of their v
- **How many:** $\operatorname{rank}(S_B) \le C - 1$, so $D' = \min(D, C-1)$
- **Regulariser:** $\tilde{S}_W = (1-\lambda) S_W + \lambda I$ stays invertible when there are more features than samples per class
- **Cost:** one eigenproblem, no gradient descent

Notes: Shao 2022 — SFDA

SFDA: the Fisher discriminant, the picture



Notes: Shao 2022 — SFDA

SFDA and H-score: Fisher, LDA and the eigenvalues

- **Fisher is LDA's direction.** For two classes $S_B \propto \Delta\Delta^\top$ with $\Delta = \mu_A - \mu_B$, so the eigenproblem gives $u \propto S_W^{-1}\Delta$, the same as LDA's $w = \Sigma^{-1}(\mu_A - \mu_B)$. LDA classifies by thresholding the projection on it
- **H-score uses the same eigenvalues.** With $\Sigma = S_W + S_B$, $S_B u = v S_W u$ rewrites as $S_B u = \frac{v}{1+v}\Sigma u$, so

$$\mathcal{H} = \operatorname{tr}\left(\Sigma^{-1} S_B\right) = \sum_i \frac{v_i}{1+v_i}, \qquad \text{each term in } [0, 1)$$

- **Check** on the H-score worked example, case A: $S_W = \operatorname{diag}(9, 0.25)$ and $S_B = \operatorname{diag}(0, 1)$, so $v = 1/0.25 = 4$ along x_2 and $\mathcal{H} = 4/(1+4) = 0.80$

Notes: Shao 2022 — SFDA, Bao 2019 — H-score

SFDA: deriving LDA from Bayes' rule

Each class is a Gaussian with a **shared** covariance, $x \mid c \sim \mathcal{N}(\mu_c, \Sigma)$, with prior q_c. Bayes' rule, then the log of the numerator expanded:

$$p(c \mid x) = \frac{q_c \mathcal{N}(x; \mu_c, \Sigma)}{\sum_k q_k \mathcal{N}(x; \mu_k, \Sigma)}, \qquad \log\left(q_c \mathcal{N}\right) = \mu_c^\top \Sigma^{-1} x - \tfrac{1}{2}\mu_c^\top \Sigma^{-1}\mu_c + \log q_c \underbrace{-\tfrac{1}{2} x^\top \Sigma^{-1} x - \tfrac{1}{2}\log|2\pi\Sigma|}_{\text{same for every class}}$$

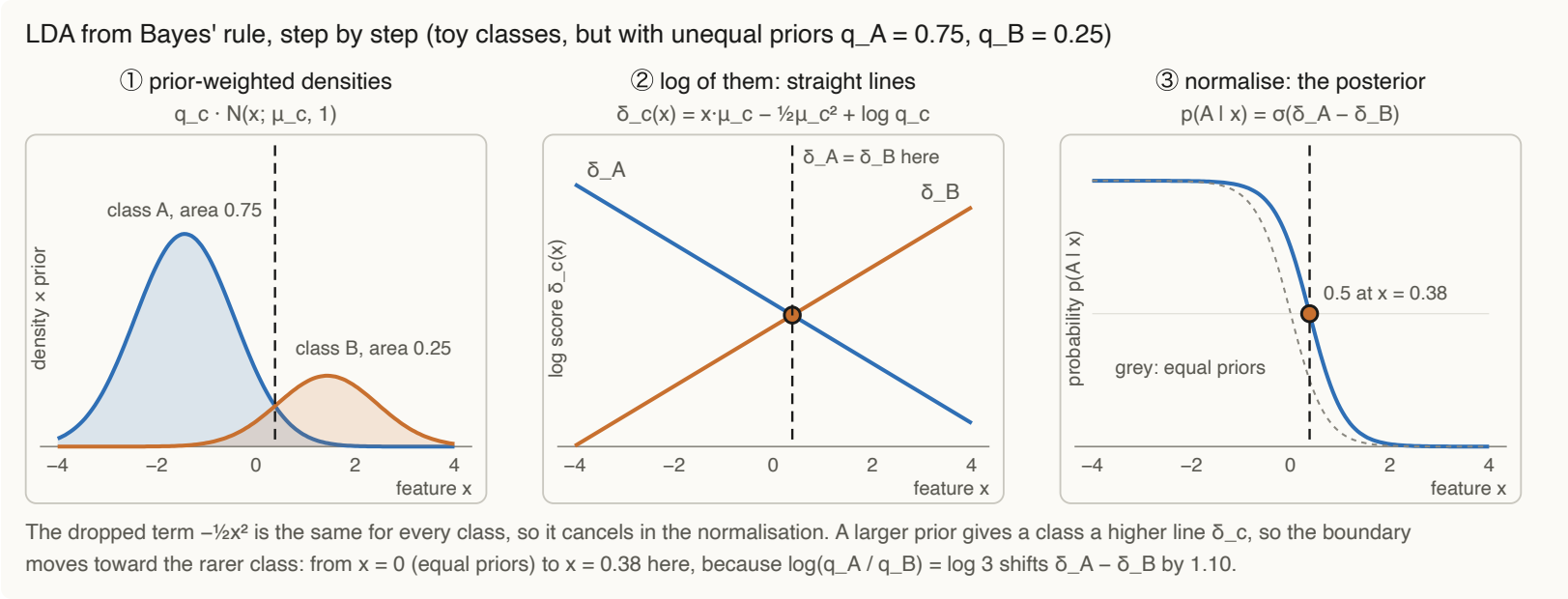
The underbraced terms multiply every term of the numerator and denominator, so they cancel:

$$\delta_c(x) = x^\top \Sigma^{-1}\mu_c - \tfrac{1}{2}\mu_c^\top \Sigma^{-1}\mu_c + \log q_c, \qquad p(c \mid x) = \frac{e^{\delta_c}}{\sum_k e^{\delta_k}}$$

- **Intuition:** $-\frac{1}{2}(x-\mu_c)^\top \Sigma^{-1}(x-\mu_c)$ is minus half the squared distance to the class mean in units of the spread, so LDA picks the nearest mean, with a head start $\log q_c$ for common classes
- **SFDA** takes $\Sigma = I$ in the Fisher space, which gives $\delta_c = x^\top \mu_c - \frac{1}{2}\|\mu_c\|^2 + \log q_c$

Notes: Shao 2022 — SFDA

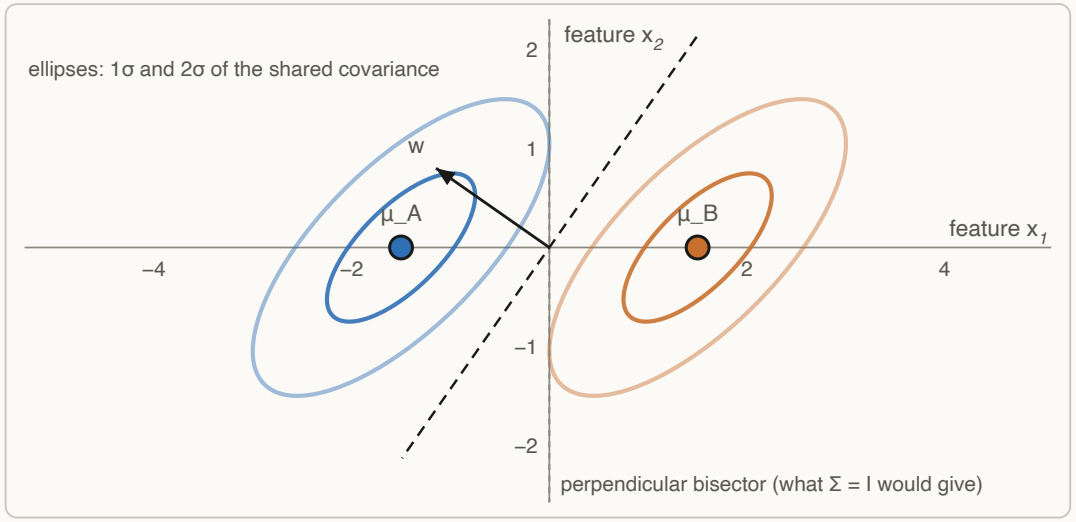
SFDA: LDA from Bayes' rule, one dimension



Notes: Shao 2022 — SFDA

SFDA: LDA with a correlated covariance

LDA in two dimensions: with one shared, correlated covariance Σ the boundary is still a straight line, tilted by Σ^{-1}



Boundary: $w^T x = 0$, $w = \Sigma^{-1}(\mu_A - \mu_B)$

$\Sigma = [[1, 0.7], [0.7, 1]]$, $\mu_A - \mu_B = (-3, 0)$
$\Sigma^{-1}(\mu_A - \mu_B) = (-5.88, 4.12)$

The classes are stretched along the diagonal, so the most reliable separating direction is not the line joining the means. Σ^{-1} corrects for the spread: it turns the boundary from the dotted vertical line to the dashed tilted one.

With Σ = I (SFDA after its Fisher projection) w is parallel to μ_A − μ_B, and the boundary is the exact perpendicular bisector.

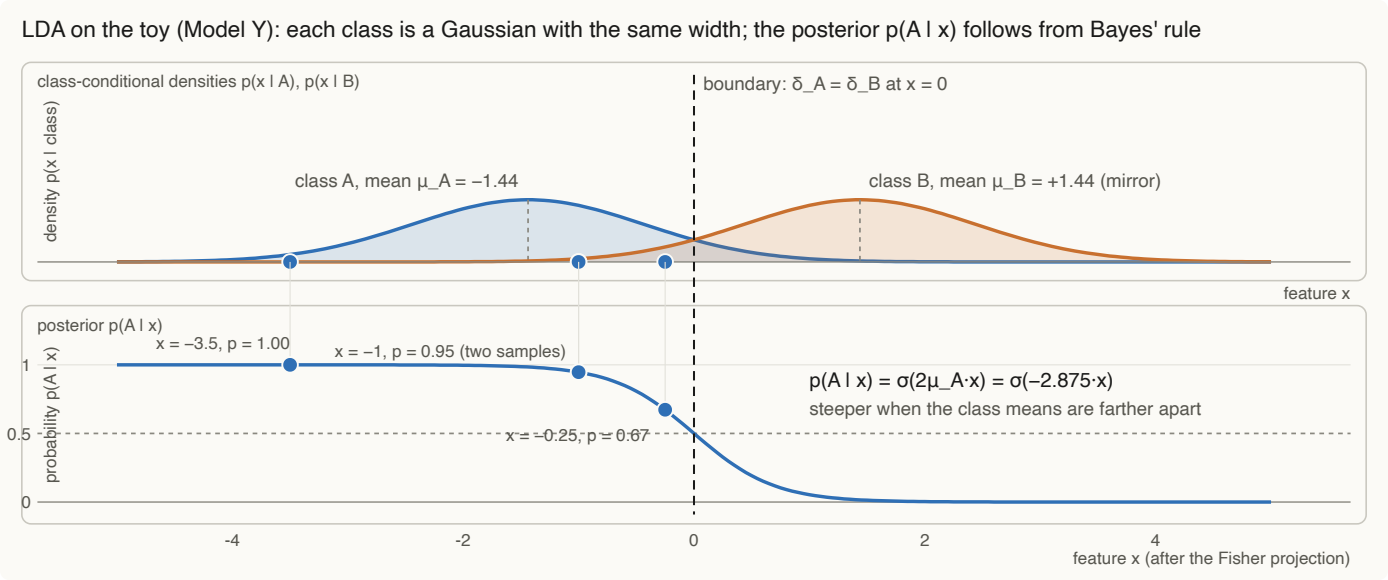
Notes: Shao 2022 — SFDA

SFDA: LDA on the toy example

LDA treats each class as a Gaussian of the same width and applies Bayes' rule. With identity covariance the $-\frac{1}{2}x^2$ term is shared by all classes and cancels:

$$\delta_c(x) = x\,\mu_c - \tfrac{1}{2}\mu_c^2 + \log q_c, \qquad p(c \mid x) = \frac{e^{\delta_c}}{\sum_k e^{\delta_k}}$$

In the toy (equal priors, $\mu_B = -\mu_A$) this reduces to $p(A \mid x) = \sigma(2\mu_A x)$: a sigmoid whose slope is the distance between the class means.



Notes: Shao 2022 — SFDA

SFDA worked example: ConfMix by hand

Binary target, so the Fisher space is 1-D ($D' = C - 1 = 1$). Take $U = 1$, equal priors, and class B the mirror image of class A, so $\mu_B = -\mu_A$ and LDA gives $p = \sigma(2\mu_A \hat{x})$ for a class-A sample. Then:

$$\tilde{x} = p\,\hat{x} + (1-p)(-\mu_A), \qquad \tilde{\mu}_A = \text{mean}(\tilde{x}), \quad \tilde{p} = \sigma(2\tilde{\mu}_A \tilde{x}), \quad \text{SFDA} = \text{mean} \log \tilde{p}$$

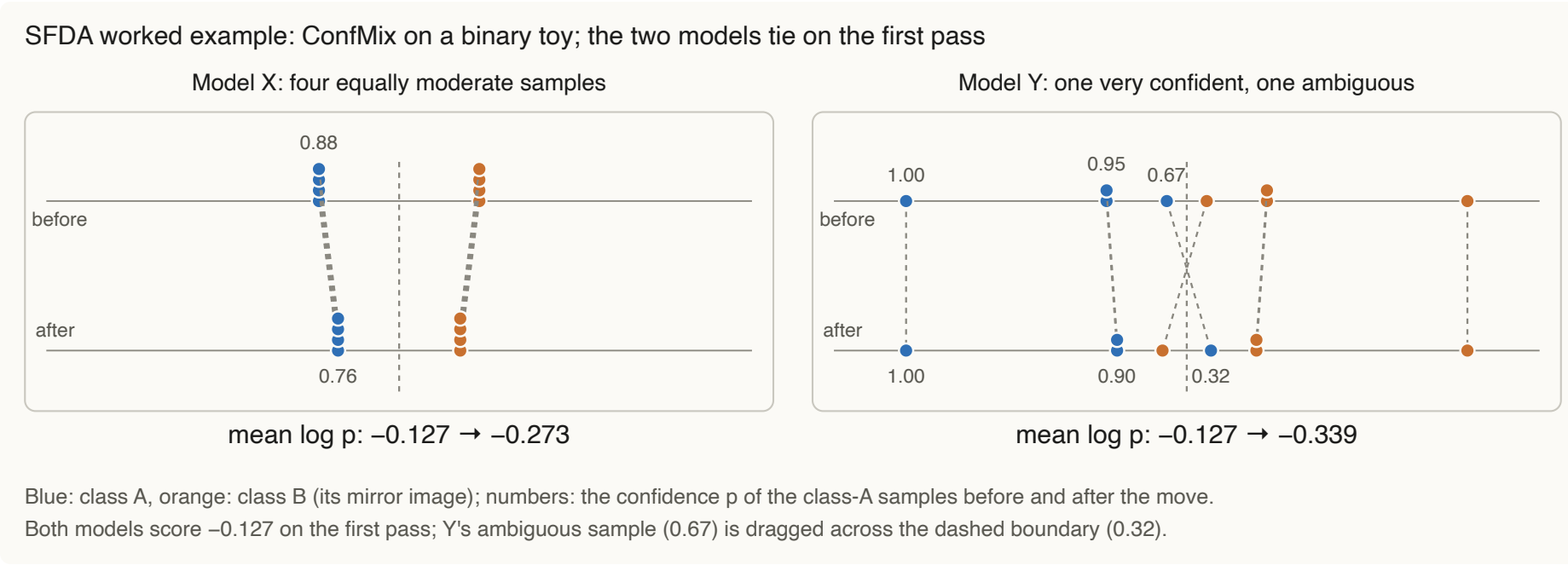
Model X ($\mu_A = -1$): four samples, all $\hat{x} = -1, p = 0.88$, so $\tilde{x} = -0.76, \tilde{p} = 0.76$. **Model Y** ($\mu_A = -1.44$, $\tilde{\mu}_A = -1.23$):

sample	$\hat{x}$	p	$\tilde{x}$	$\tilde{p}$
1	-3.50	1.00	-3.50	1.00
2, 4	-1.00	0.95	-0.87	0.90
3	-0.25	0.67	$+0.30$	0.32
$\text{mean} \log p$		-0.127		-0.339

Model X goes from -0.127 to -0.273. Both models tie on the first pass; ConfMix drags Y's ambiguous sample 3 across the boundary (its confidence falls from 0.67 to 0.32), and Y ends lower. A constructed toy, to show the mechanism.

Notes: Shao 2022 — SFDA

SFDA worked example: the picture



Notes: Shao 2022 — SFDA